



COEP Technological University

(COEP Tech)

A Unitary Public University of Government of Maharashtra

w.e.f 21st June 2022

(Formerly College of Engineering Pune)

YEAR : SECOND YEAR

SEMISTER: III

SUBJECT: Geo-informatics

Prepared by,

Jaysing Jadhav

Adjunct Faculty, Planning section,

Dept. of Civil Engineering CoEP

Introduction

- The Land Suitability Analysis (LSA) project is a GIS-based process for evaluating the suitability of land for development.
- The two major outputs of the LSA project are an environmental composite map and a land suitability map.
- The environmental composite map shows the extent and overlap of natural features and environmental conditions that indicate the capability and limitations of natural systems for urban development.
- The land suitability map shows the relative suitability of land in a planning area for urban-type development.

- Land suitability analysis is a mandatory component of the local land use plan.
- It is a process for determining a planning area's supply of land that is suitable for development.
- The analysis includes consideration of a number of factors, including natural system constraints, compatibility with existing land uses and development patterns, existing land use policies, and the availability of community facilities.
- A key output of the analysis is a land suitability map that shows vacant or under-utilized land that is suited for the development.
- This map is a major part of the foundation for the development of local land use policies and the future land use map.

GIS Approach

- Land suitability analysis involves the application of criteria to the landscape to assess where land is most and least suitable for development of structures and infrastructure.
- The system enables planners to create and modify a land suitability analysis that makes the best use of available data.

GIS Tools for Land Suitability Analysis

i. **GIS and Spatial Analysis**

- In addition to storing, retrieving, displaying spatial data, a geographic information system enables the user to create buffers, overlays, intersections, proximity analysis, spatial joins, map algebra, and other analytical operations.
- In the context of land suitability, GIS helps the user determine what locations are most/least suitable for development.
- In this way, the results of GIS analysis can provide support for decision-making.

- The eight steps in Spatial Analysis include:
 - a) Define criteria for the analysis
 - b) Define data needed
 - c) Determine what GIS analysis operations should be performed
 - d) Prepare the data
 - e) Create a model
 - f) Run the model
 - g) Analyze results
 - h) Refine the model as needed

ii. Raster vs. Vector Approach

- There are two possible data models that can be used in a GIS: vector and raster.
- The raster data model represents features as a matrix of cells (pixels) in continuous space.
- Vector data consist of discrete points, lines, and polygons. These feature shapes are defined by x and y coordinates.
- Raster data are used for land suitability modeling because analysis can be performed on several raster layers at once.
- For example, raster data enable the user to perform a weighted overlay on several layers. Vector data enable analysis on only two layers at a time in an operation that requires a great deal of computer resources.
- Raster data provide continuous coverage of a geographic area and analysis is much more efficient.

Technical Issues with a Raster Data Model

Resolution

- The cell size used for a raster layer will affect the results of the analysis and how the map looks.
- The cell size should be based on the original map scale and the minimum mapping unit.
- Using too large a cell size will cause some information to be lost.
- Using a cell size that is too small requires a lot of storage space, and takes longer to process, without adding additional precision to the map.
- For a given analysis, you will need to decide the optimal resolution to maximize accuracy and performance.
- The higher the resolution, the greater the accuracy; but performance suffers.

Pixels contain one value only

- Limiting a cell to one value can misrepresent spatial data.
- For example, the boundary of two soil types may run across the middle of a cell.
- In such cases, the cell is given the value of the largest fraction of the cell, or the value of the middle point in the cell.

Only one item of information is available for each location within a single layer

Multiple items of information require multiple layers.

If, in a soils vector layer, you have two attributes—septic suitability and flood frequency--you will have to create two raster layers: one that contains septic suitability information and one that contains flood frequency information.

Discrete and continuous phenomena

- Discrete phenomena

- ✓ Landuse
- ✓ Ownership
- ✓ Political boundaries

0	1	1	2
No Data	1	1	1
No Data	1	2	2
1	1	2	2

Vegetation
0 = Barren
1 = Forest
2 = Water

- Continuous phenomena

- ✓ Elevation
- ✓ Distance
- ✓ Density
- ✓ Suitability

1.12	1.75	1.81	2.03
0.26	1.63	1.87	1.98
0.00	0.91	0.73	1.42
0.00	0.18	No Data	No Data

Rainfall
(inches)

iii. Introduction to Spatial Analyst

- The ESRI Spatial Analyst extension enables the user to create, query, map, and analyze cell-based raster data and to perform integrated vector–raster analysis.
- Spatial Analyst enables desktop GIS users to create, query, and analyze cell-based raster maps; derive new information from existing data; query information across multiple data layers; fully integrate cell-based raster data with traditional vector data sources; and create sophisticated spatial models using ModelBuilder.
- For the Land Suitability Analysis, users can rate areas according to several factors with varying weights and values, and derive new information from existing data to determine land suitability.

- Additional capabilities available through the standard user interface include queries on multiple grid themes, neighborhood and zone analysis, grid classification and display, summary histograms, and more.
- Operations available with Spatial Analyst:
 - ✓ Convert feature themes (point, line, or polygon) to grids
 - ✓ Create raster buffers based on distance from any raster or vector feature
 - ✓ Create density maps of point features
 - ✓ Perform Boolean queries and algebraic calculations on multiple grid themes simultaneously
 - ✓ Do neighborhood and zone analysis
 - ✓ Display and reclassify grid data

iv. Introduction To Model Builder

ModelBuilder is a tool for creating and managing automated and self-documenting spatial models.

Modelbuilder enables users to create process- flow diagrams and scenarios to automate the modeling process.

Users can easily change the data sets used by the model, modify the influence of each data set on the model, perform complex analysis functions, and generate maps that illustrate the results of analysis.

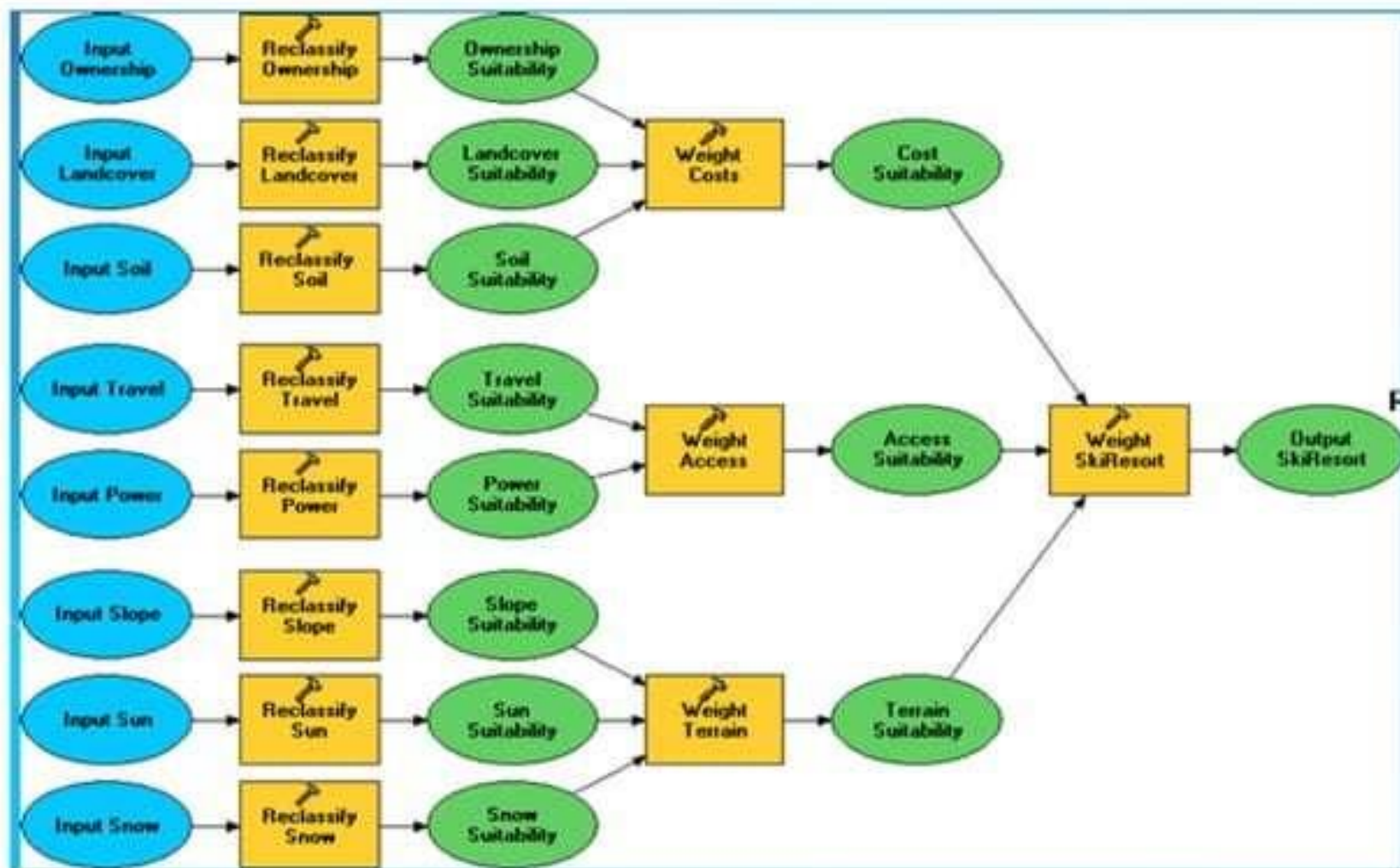
Data derived from one model can be used as input for another model.

Users can run a model with a variety of parameters to assess data sensitivity or to evaluate geographically different but structurally similar data sets.

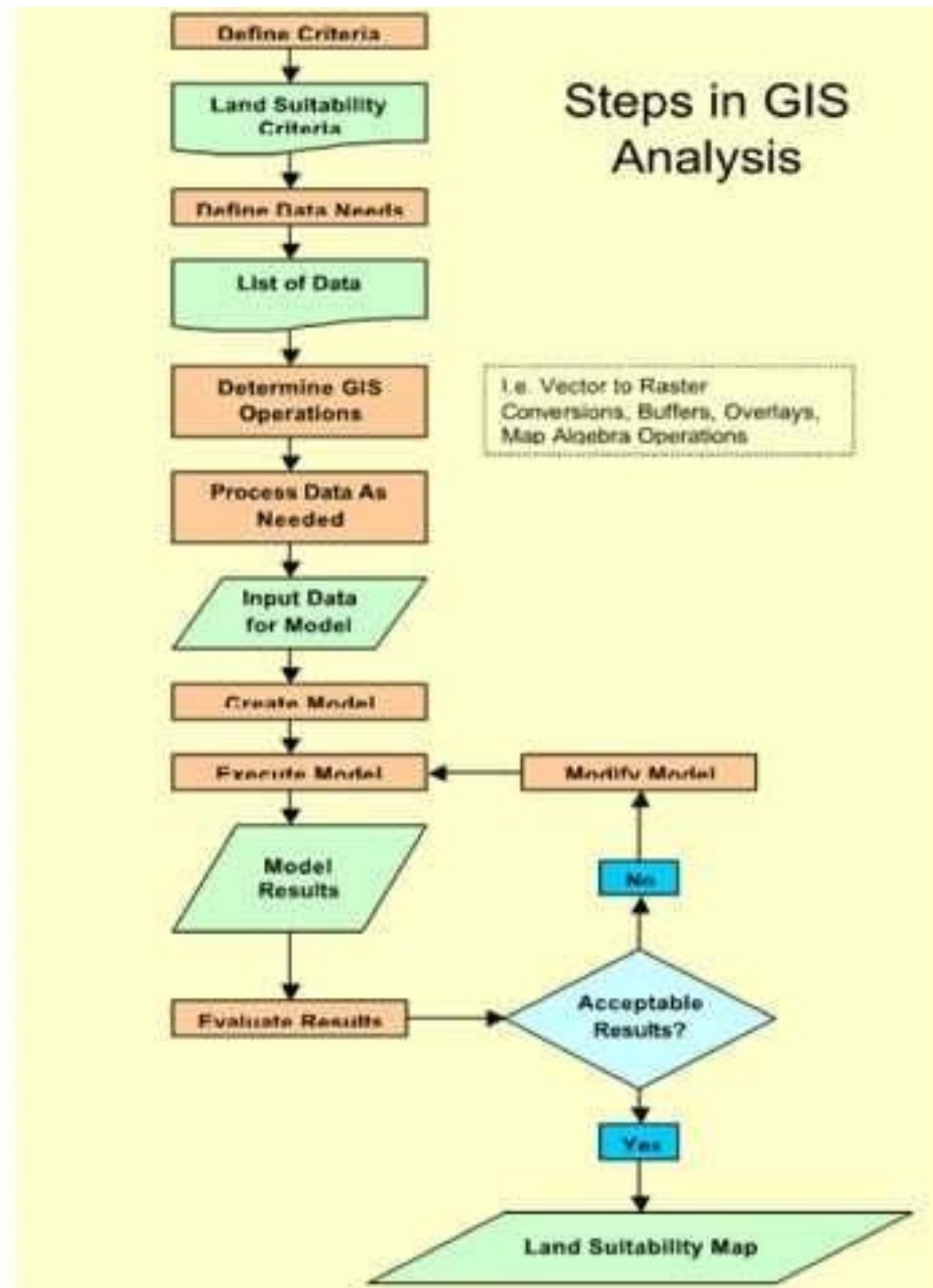
Users can copy portions of their models within a model and smaller models can be combined to build larger models.

- In the case of Land Suitability Analysis, the layer weights can be easily changed, and the models may be re-run to evaluate the new results.
- ModelBuilder is ideal for this task because it allows users to overlay multiple layers, rank order categories within each layer, include a weight for each layer, and sum using map algebra.
- ModelBuilder creates a process-flow diagram that displays the layers and operations.
- For example, the land suitability model combines and classifies multiple GIS layers to produce a land suitability map as illustrated in the figure below.

Model Builder Example



Land Suitability Model



Define the Criteria

- criteria for the Land Suitability Analysis is based on the Guidelines set and modified criteria according to available datasets.
- The criteria for suitability for development (high, medium, low, and least suitable) can be identified as follows:
- Within 100-year **Flood Zones** have low development suitability

- Within **Water Supply Watersheds** have low suitability
- Within 500 feet of a Significant **Natural Heritage Area** have low suitability
- Within 500 feet of a **Hazardous Substance Disposal Site** have low suitability
- Within 500 feet of a **Wastewater Treatment Plant** have low suitability
- Within 500 feet of a **Municipal Sewage Discharge Point** have low suitability
- Within 500 feet of a **Land Application Site** have low suitability
- Within 500 feet of **an Airport** have low suitability
- Within a half-mile of **Primary Roads** have high suitability; within a half-mile to a mile have medium suitability; areas greater than one mile outside of primary roads have low suitability
- Within a half-mile of **Developed Land** have high suitability; areas within a half-mile to a mile have medium suitability; areas further than one mile away from developed land have low suitability

- Within a quarter-mile of **Water Pipes** have high suitability; areas within a quarter-mile to a half-mile of water pipes have medium suitability; areas further than a half-mile away from water pipes have low suitability
- Within a quarter-mile of **Sewer Pipes** have high suitability; areas within a quarter-mile to a half-mile of sewer pipes have medium suitability; areas further than a half-mile away from water pipes have low suitability
- Within **government reserves or State Lands** are LEAST suitable
- Within **Protected Lands** are LEAST suitable
- Within **Estuarine Waters** are LEAST suitable

- According to these criteria, values for layers are quantitatively scored according to suitability for development.
- For example, an area that is inside a storm surge area or within 500 feet of a Significant Natural Heritage Area has low suitability. These areas receive a score of -2 (negative 2).
- An area that is close to existing infrastructure (roads, sewer lines, existing development, etc.) has high suitability for development. These areas receive a score of +2 (positive two).
- Note that the proximity concept is represented by a buffer in the model. A buffer should not be smaller than the distance of one side of a cell. In this case, the smallest buffer is 500 feet and a cell has a width of 209 feet.
- Also, to account for proximity of features to cells on the boundaries of the study area (county), themes that are subject to buffers are clipped to a polygon of the county plus 2 miles (2-mile buffer of county boundary including the county).
- The final map will be clipped to the county boundary (not buffered).

- Additionally, most the data layers are ranked according to how important they are to the overall analysis.
- In the criteria spreadsheet developed in the Table below, users may rank a layer as 1, 2 or 3, with 3 being very important.
- Other values may be used, but keep in mind the advantage of keeping the factors relatively uncomplicated for presentation and explanation in public meetings.
- The least suitable areas (protected lands, military areas, coastal wetlands, estuarine waters, and exceptional and substantial non-coastal wetlands) are treated somewhat differently.
- They are given scores of 0 or 1. Areas within protected lands, coastal wetlands, etc., receive a score of 0. Areas outside of these sensitive areas receive a score of 1.

Criteria Table Example

Note that the first set of layers (green shading) are either least suitable (the value of zero will be multiplied by the results of the layers with white and gray shading for a product of zero) or medium suitability (the value of one will be multiplied by the results of the other sets of layers for a product equal to the score based on those other sets).

	-----Criteria and Rating-----			
Layer Name	Least Suitable	Low Suitability	Medium Suitability	High Suitability
	0	-2	1	2
Coastal Wetlands	Inside		Outside	
Exceptional and Substantial Noncoastal Wetlands	Inside		Outside	
Estuarine Waters	Inside		Outside	
Protected Lands	Inside		Outside	
Beneficial Noncoastal Wetlands		Inside		Outside
Storm Surge Areas		Inside		Outside
Soils with septic limitations		Severe	Moderate	Slight
Flood Zones		Inside		Outside
Water Supply Watersheds		Inside		Outside
Significant Natural Heritage Areas		< 500'		> 500'
Hazardous Substance Disposal Sites		< 500'		> 500'
NPDES Sites		< 500'		> 500'
Wastewater Treatment Plants		< 500'		> 500'
Municipal Sewer Discharge Points		< 500'		> 500'
Airports		< 500'		> 500'
Developed Land		> 1 mi	.5 - 1 mi	< .5 mi
Primary Roads		> 1 mi	.5 - 1 mi	< .5 mi
Water Pipes		> .5 mi	.25 - .5 mi	< .25 mi
Sewer Pipes		> .5 mi	.25 - .5 mi	< .25 mi
Total				

- The next step is to rank the layers from 1 to 3 with 3 representing the most weight in land suitability.
- The spreadsheet included on the Land Suitability CD is ready for the user to modify the default weights (*see the next Table*).
- Once a ranking is agreed upon, the model requires that the user quantify the ranked layers from an ordinal scale (ranked 1 thru 3) to a percentage of the total (percent weight) to assign relative weights.
- The relative weight for a layer is equal to 100 (percent) divided by the product of the sum of all rankings times the ranking for that layer.
- In other words, it is the whole pie divided by the number of pieces (yielding the size of a piece), times the number of pieces for that layer.
- If all layers were assigned a weight of 1, the relative weight in percent for any one layer would be equal to the 100 divided by the number of layers (one equal piece of pie each).
- The far right column of the spreadsheet expresses the relative weights as a ratio (or “multiplier” required for the model, below).
- Note that these numbers change for each county depending on the number of layers that apply. The calculations are already set in formulas in the spreadsheet.

Example of Rankings and Percent Weights

Layer Name	Criteria and Rating				Assigned Weight	Percent Weight	Multiplier
	Least Suitable	Low Suitability	Medium Suitability	High Suitability			
	0	-2	1	2			
Coastal Wetlands	Inside		Outside				
Exceptional and Substantial Noncoastal Wetlands	Inside		Outside				
Estuarine Waters	Inside		Outside				
Protected Lands	Inside		Outside				
Beneficial Noncoastal Wetlands		Inside		Outside	1	4.348	0.04348
Storm Surge Areas		Inside		Outside	2	8.696	0.08696
Soils with septic limitations		Severe	Moderate	Slight	1	4.348	0.04348
Flood Zones		Inside		Outside	2	8.696	0.08696
Water Supply Watersheds		Inside		Outside	1	4.348	0.04348
Significant Natural Heritage Areas		< 500'		> 500'	2	8.696	0.08696
Hazardous Substance Disposal Sites		< 500'		> 500'	1	4.348	0.04348
NPDES Sites		< 500'		> 500'	1	4.348	0.04348
Wastewater Treatment Plants		< 500'		> 500'	1	4.348	0.04348
Municipal Sewer Discharge Points		< 500'		> 500'	1	4.348	0.04348
Airports		< 500'		> 500'	1	4.348	0.04348
Developed Land		> 1 mi	.5 - 1 mi	< .5 mi	1	4.348	0.04348
Primary Roads		> 1 mi	.5 - 1 mi	< .5 mi	2	8.696	0.08696
Water Pipes		> .5 mi	.25 - .5 mi	< .25 mi	3	13.043	0.13043
Sewer Pipes		> .5 mi	.25 - .5 mi	< .25 mi	3	13.043	0.13043
Total					23	100.000	1.00000

Define the Data

- The final determination of the factors included in the analysis is influenced by the availability of digital data layers.
- The data are projected to one datum in order to have similar coordinate system

Determine the GIS Operations

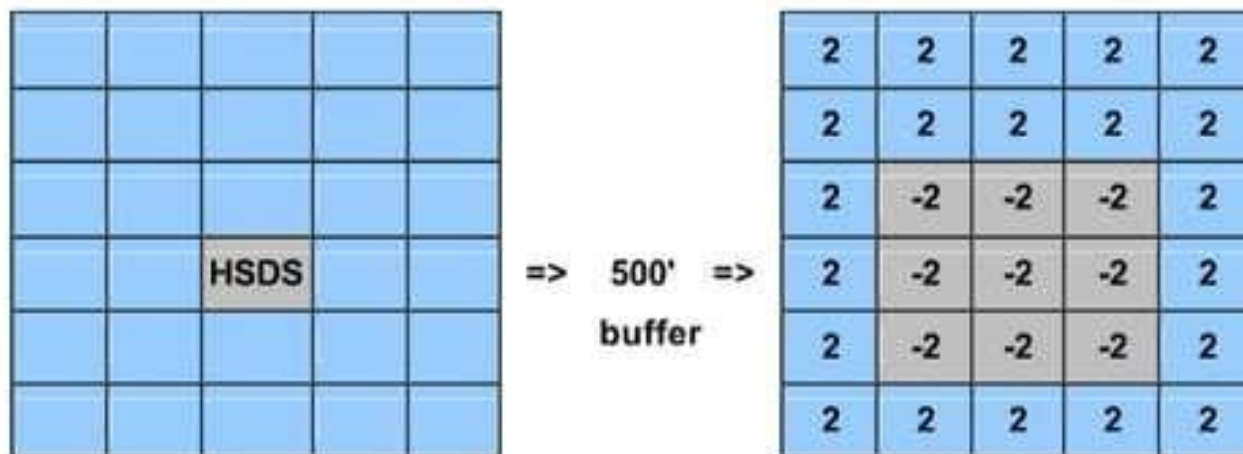
- Based on the established criteria and data, the next step is to define what operations need to be performed in order to determine land suitability.
- Many layers will have to be converted from vector to raster.
- Once in raster format, each layer's values need to be reclassified into either the 1's and 0's scoring system, or the -2 thru +2 scoring system.
- Buffering will have to be done on many layers to determine what values should be assigned inside/outside the extent of the feature and its buffer.
- For example, airports are buffered by 500 feet. Any areas within that buffer are assigned a value of -2; areas outside are assigned a value of +2.

Operations used in this analysis:

- Raster to Vector Conversion
- Buffer
- Reclassification
- Map Algebra – multiply by a constant (absolute weight)
- Map Algebra – add multiple layers
- Map Algebra – multiply layers

- **Vector to Raster Conversion:**
- Layers must be converted from vector to raster to be used in the model.
- Some layers are converted within the model itself. Others have already been converted outside of the model.

- **Buffer:**
- Many criteria specify that areas within a specific feature have suitability; outside have high suitability (and vice-versa).
- Example: Areas within 500 feet of a Hazardous Substance Disposal Site have low suitability.



Example: Areas within 500 feet of a Hazardous Substance Disposal Site have low suitability.

- **Reclassify:**
- Some layers need to be reclassified. For example, the 'Soils With Septic Limitations' layer has a 'septic' attribute that contains values, Severe, Moderate, or Slight.
- These values are reclassified to -2, 1, and +2 respectively.

Mod	Sev	Sev	Sev	Slight	Mod		1	-2	-2	-2	+2	1
Mod	Mod	Sev	Sev	Mod	Mod		1	1	-2	-2	1	1
Mod	Mod	Mod	Slight	Slight	Slight	-> Reclassify ->	1	1	1	+2	+2	+2
Slight	Mod	Mod	Mod	Slight	Slight		+2	1	1	1	+2	+2
Slight	Mod	Sev	Sev	Mod	Slight		+2	1	-2	-2	1	+2
Slight	Slight	Mod	Sev	Sev	Slight		+2	+2	1	-2	-2	+2

- **Map Algebra**
- Multiply by a constant: The weighted layers will each be multiplied by their respective absolute weight. For example, all the values in the Storm Surge Areas will be multiplied by 0.08696 assuming the criteria listed above

1	-2	-2	-2	+2	1
1	1	-2	-2	1	1
1	1	1	+2	+2	+2
+2	1	1	1	+2	+2
+2	1	-2	-2	1	+2
+2	+2	1	-2	-2	+2

X 0.08696 =

0.09	-0.2	-0.2	-0.2	0.17	0.09
0.09	0.09	-0.2	-0.2	0.09	0.09
0.09	0.09	0.09	0.17	0.17	0.17
0.17	0.09	0.09	0.09	0.17	0.17
0.17	0.09	-0.2	-0.2	0.09	0.17
0.17	0.17	0.09	-0.2	-0.2	0.17

- **Map Algebra – Add Multiple Layers:**
- After all weighted layers are multiplied by their respective constants, they will be added together to get a suitability rating. The following example shows only two of the layers being added (allow for rounding in addition). When all layers are added, the resultant layer has values from – 2 to +2.

0.09	-0.2	-0.2	-0.2	0.17	0.09		-0.1	-0.1	0.09	0.09	0.09	0.09		0	-0.3	-0.1	-0.1	0.26	0.17
0.09	0.09	-0.2	-0.2	0.09	0.09		-0.1	-0.1	-0.1	0.09	0.09	0.09		0	0	-0.3	-0.1	0.17	0.17
0.09	0.09	0.09	0.17	0.17	0.17	+	-0.1	-0.1	-0.1	-0.1	0.09	0.09	=	0	0	0	0.09	0.26	0.26
0.17	0.09	0.09	0.09	0.17	0.17		-0.1	-0.1	-0.1	-0.1	-0.1	0.09		0.09	0	0	0	0.09	0.26
0.17	0.09	-0.2	-0.2	0.09	0.17		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.09	0	-0.3	-0.3	0	0.09
0.17	0.17	0.09	-0.2	-0.2	0.17		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.09	0.09	0	-0.3	-0.3	0.09

- Map Algebra – Multiply Layers:** Layers that have features to be scored least suitable are classified with 0's and 1's, then the layers multiplied together. The resulting layer shows all areas least suitable for development.

Military Areas		Coastal Wetlands		Least Suitable
1 1 1 1 1		1 1 0 0 1		1 1 0 0 1
1 0 0 1 1		1 1 0 0 1		1 0 0 0 1
1 0 0 1 1	x	1 1 1 0 1	=	1 0 0 0 1
1 0 0 1 1		1 1 1 1 1		1 0 0 1 1
1 1 1 1 1		1 0 0 0 1		1 0 0 0 1
1 1 1 1 1		1 0 0 1 1		1 0 0 1 1

County Boundary		LSA Map		Clipped LSA
1 1 1 1 1		1 2 2 1 2		2 2 2 2
1 1 1 1 1		1 1 2 2 2		1 2 2 2
1 1 1 1 1	x	2 2 3 2 2	=	2 3 2 2
1 1 1 1 1		3 3 3 2 1		3 3 3 2 1
1 1 1 1 1		5 4 4 4 2		4 4 4 4
1 1 1 1 1		5 2 4 4 3		4 4

Multiply the reclassified county boundary with the land suitability. The No Data values will drop out, clipping the land suitability map.

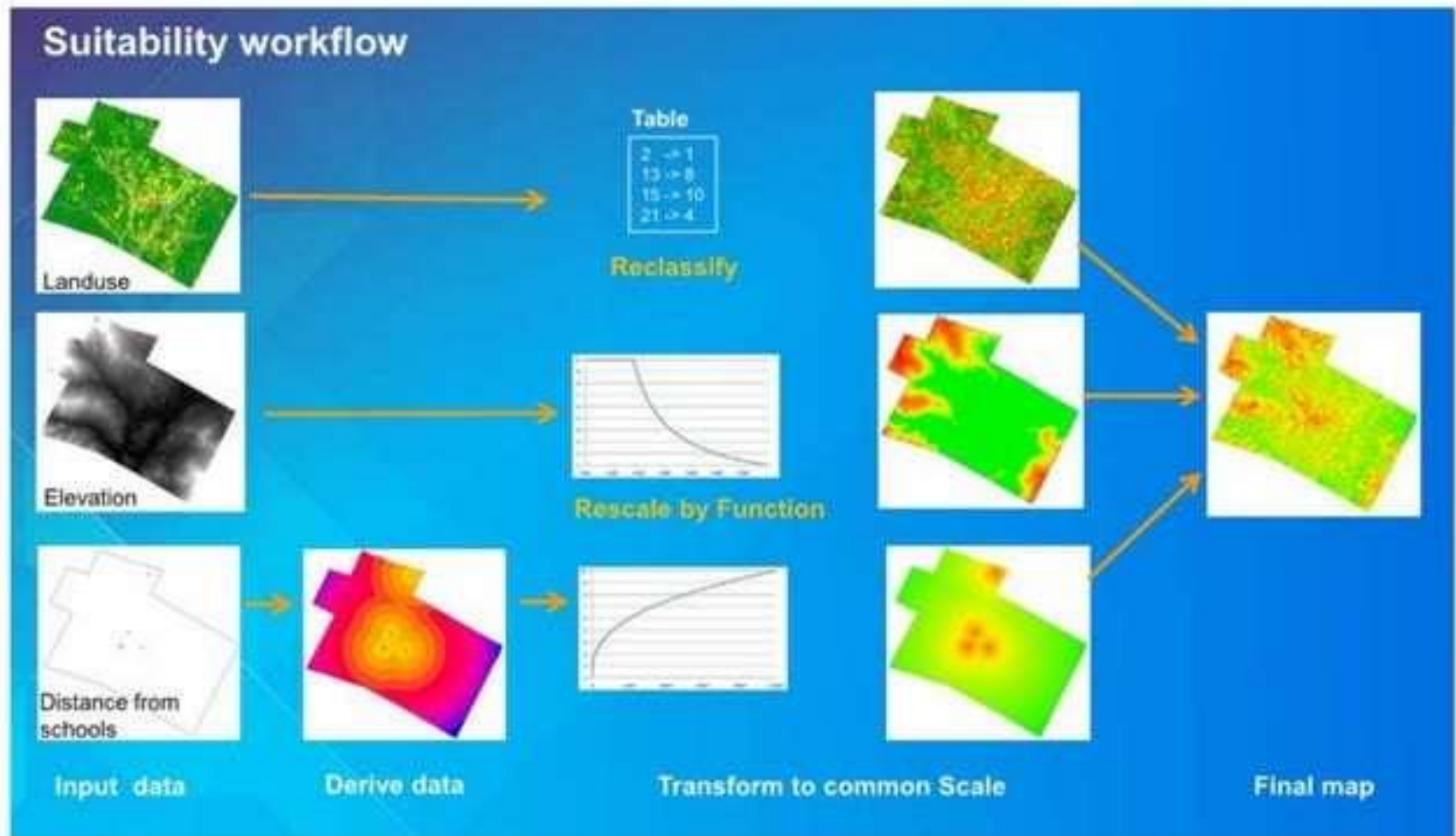
Data Preparation

- After the GIS operations are determined, the data must be prepared for the Model.
- This includes clipping the data to the correct boundary; creating subsets of data such as coastal wetlands versus all wetlands; and even converting some data to raster format before it is added to the model.

Using ModelBuilder

- Map document files (.mxd) are provided for the ArcGIS models. The model is located in a new Toolbox in ArcToolbox (LSA Model and Environmental Composite Model). To run or make edits to the model – double click the Toolbox function and right click the model and select Edit. A new window will open. This is the Model Builder interface.

Running the Land Suitability Model



Evaluating the Results

- ArcMap permits the classification of results by natural breaks, equal intervals, etc.
- Natural breaks is a better classification of the results.
- Natural breaks are best for comparing relative suitability of resources within specific planning area within a county
- Equal intervals may mask subtle differences between suitability of locations within the planning area
- Equal interval may mean that some areas have few or no areas suitable for development
- Significant research may be required to determine ranges or the ranges would be more arbitrary than natural breaks
- Natural breaks appear to be more statistically valid than equal intervals

- Verify the results by viewing the newly classified grid underneath the vector layers.
- The land suitability pattern should be related to vector layers visually, though of course the model has computed the spatial relationships in a way that the vector layers cannot.

Environmental Composite Map

- This map show the location of three categories of land based on natural features and environmental conditions
 1. **Class I** is land that contains only minimal hazards and limitations which can be addressed by commonly accepted land planning and development practices. Class I land will generally support the more intensive types of land uses and development.
 2. **Class II** is land that has hazards and limitations for development that can be addressed by restrictions on land uses, special site planning, or the provision of public services, such as water and sewer. Land in this class will generally support only the less intensive uses, such as low density residential, without significant investment in services.
 3. **Class III** is land that has serious hazards and limitations. Land in this class will generally support very low intensity uses, such as conservation and open space.

Environmental Criteria

Layer		Class I	Class II	Class III
Coastal Wetlands				✓
Exceptional or Substantial Non-Coastal Wetlands				✓
Beneficial Non-Coastal Wetlands			✓	
Estuarine Waters				✓
Soils with Slight or Moderate Septic Limitations		✓		
Soils with Severe Septic Limitations				✓
Flood Zones			✓	
Storm Surge Areas			✓	
HQW/ORW Watersheds			✓	
Water Supply Watersheds			✓	
Significant Natural Heritage Areas			✓	
Protected Lands				✓

- For a given cell, the computed value of the cell will be determined by the highest class theme that contains the cell.
- For example, if a cell is in a coastal wetland (Class III) and in a storm surge area (Class II) and intersects a soil with a slight or moderate septic limitation (Class I), the cell value will be Class III.
- In other words, if a cell does not meet the criteria for Class III, but qualifies as Class II, it has Class II for a value.
- If a cell does not qualify for either Class III or Class II, then it may be Class I or contain no data from the themes identified in the criteria.

- The resulting Environmental Composite Map is similar to the Land Suitability Map in that Class III areas are consistent with the Least Suitable category and the Class I areas are related to the Most Suitable areas.
- The primary difference is the absence of infrastructure in the Environmental Composite Map that heightens the emphasis on environmental sensitivity and relative land conservation value.

Thank You